P-192 product quotient: marked action and polarization

P-192 product quotient by the known generator (G,Q)
Unpolarized quotient is always isomorphic to E squared

Retain quotient marking and type-(1,r) polarization
Local line stabilizer has additive index r

Non-scalar marked action -> quadratic scalar equation
Construction of that action remains the algorithmic target

Derived geometric construction. Let E be P-192 over F_p with E(F_p)=<G> of prime order r, Q=xG, and pi the p-power Frobenius. Put alpha=pi-1. Its norm is p+1-t=r, its differential is -1, and ker(alpha)=E(F_p). The matrix F_x=[[alpha,0],[-x,1]] defines an F_p-rational separable isogeny E^2 -> E^2 of degree r. Its kernel is exactly Gamma_x=<(G,Q)>. Thus A_x=E^2/Gamma_x is isomorphic to E^2 as an unpolarized abelian variety, for every x. The displayed isogeny uses x and is an existence argument, not a way to construct it from unknown Q. [pcc:unverified]

Proof. F_x(P,R)=0 means alpha(P)=0 and R=xP. Since all of ker(alpha) is <G>, the kernel is Gamma_x. The matrix factors as diag(alpha,1) times the determinant-one integer automorphism [[1,0],[-x,1]], giving degree r and separability. Factoring through the quotient gives an isomorphism. Consequently the abstract unpolarized endomorphism ring is M_2(O), independent of x, where O=End(E). Computing only this abstract ring or the quotient isomorphism class does not recover the marked scalar. [pcc:unverified]

Marked action extractor. An integral matrix M=[[a,b],[c,d]] preserves Gamma_x precisely when M(1,x) is a multiple of (1,x) modulo r. Eliminating that multiple gives b*x^2+(a-d)*x-c=0. If this polynomial is nonzero modulo r, factor it and check the at most two candidates against [x]G=Q. For CM entries, first apply the known character chi:O -> F_r describing action on G. In the certified P-192 basis O=Z[pi], chi(u+v*pi)=u+v mod r. This gives the same equation with chi-reduced entries. [pcc:unverified]

The local marking that must be retained. Since O/(alpha)=F_r, reduction maps M_2(O) onto M_2(F_r). The line stabilizer has r^3 matrices among r^4, hence its inverse image S_x has additive index r in M_2(O). It is a subring, conjugate modulo alpha to the upper triangular matrices. The embedded order is S_x=M_2(O) intersect F_x^(-1)M_2(O)F_x. There are r+1 lines in the two-dimensional residue space; this line marking is additional to the imaginary quadratic field and its class number. This is an index calculation, not a runtime lower bound or a class-number formula for a surface. [pcc:unverified]

A conditionally constructive route. Build a usable model of the marked quotient from the known generator (G,Q); compute endomorphisms with their action transported through that quotient map; intersect with integral source matrices; obtain a matrix whose reduced action is not scalar; factor the displayed quadratic and verify candidates. Each of the first three steps requires an algorithm and an accounted cost. An abstract endomorphism basis with no transport data is insufficient. Clearing denominators divisible by alpha can erase the reduced action and must not be treated as a free conversion. The prime quotient degree r has no nontrivial factorization into smaller-degree isogenies in a direct chain, although that alone does not exclude an auxiliary higher-dimensional construction. [pcc:unverified]

Polarization retained explicitly. In the coordinates supplied by the x-dependent isomorphism F_x, let H_x=[[1+x^2,x*alpha],[x*bar(alpha),r]]. This is an integral Hermitian positive matrix with determinant r, and F_x^dagger H_x F_x=rI. It represents the descended polarization of type (1,r), rather than a principal polarization. Thus the unpolarized quotient collapse does not justify dropping polarization or the quotient marking. The formula is conditional on x; it is not an unknown-secret construction of H_x. The exact CM arithmetic checker verifies the pullback identity and determinant for every fixture. [pcc:unverified]

Point-action controls. Five public secrets, including two full-width SHA256-derived scalars, produce index-r lattices of triples (u,b,c) satisfying u*x+b*x^2-c=0. The producer applies LLL and chooses a short row, balances the diagonal entries a-d=u, evaluates the resulting matrix on (G,Q), and extracts candidate roots. A separate checker verifies the lattice determinant, congruences, actual point action, roots and final scalar check. The two full-width fixtures produce 63-bit maximum matrix entries; no optimality, distribution or unknown-secret construction claim follows. These lattices were built using their public secret. [pcc:unverified]

Finite local-ring control. For each of 11 finite slopes over F_11, exhaustive matrix enumeration finds 1331 line stabilizers, of which 1100 are invertible. The checker independently constructs them by conjugating upper triangular matrices. This

checks all finite slopes of that toy field; the general index-r statement follows from the conjugation argument. The infinity slope is covered by the algebraic argument and was not enumerated. [pcc:unverified]

Failure record. The first producer invocation used unsupported one-index row access in python-flint and raised TypeError before producing a certificate. The log is preserved; two-index access fixed the interface error. The completed point, lattice, local-ring and polarization checks pass. No cloud job was submitted.

Literature boundary. Jetchev and Wesolowski study horizontal isogeny graphs and random self-reducibility for suitable ordinary abelian varieties: https://arxiv.org/abs/1506.00522 . Brooks, Jetchev and Wesolowski explicitly study absolutely simple ordinary varieties and principally polarizable surface graphs: https://arxiv.org/abs/1609.09793 . The split product here has algebra M_2(K), and the displayed descended polarization has type (1,r); those results cannot simply be imported as a cost bound for this marked construction. The accessible abstracts were checked; two IACR PDF requests returned HTTP 403.

Next construction. Determine whether the marked polarization or a suitable auxiliary prime-to-r isogeny construction can expose a non-scalar integral stabilizer without enumerating the order-r kernel. Any proposed use of the smaller CM class number must preserve the local line action through that construction. The unpolarized quotient alone is now ruled out as the distinguishing object. [pcc:unverified]